



ROOT



MATLAB Geek

Summer Bootcamp | 2026

WiseDuck



Code is poetry written for
machines!

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Let's Code!



The Wanderer



EXAMPLE 4,0 | ODE

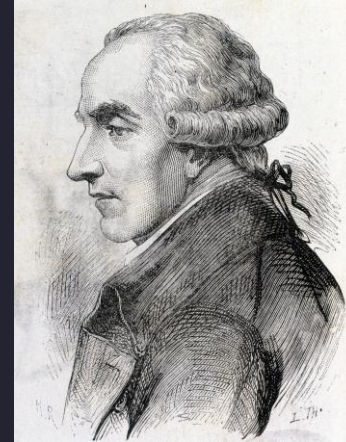
Solve the following ordinary differential equation.

$$\begin{cases} \frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 10y = 0 \\ y(0) = 1 \\ y'(0) = 0 \end{cases}$$



EXAMPLE 4,1 | Laplace

$f(t)$	$F(s)$
a	$\frac{a}{s}$
e^{at}	$\frac{1}{s - a}$
$t^n \quad (n \in \mathbb{Z})$	$\frac{n!}{s^{n+1}}$
$t^n \quad (n \notin \mathbb{Z})$	$\frac{\Gamma(n + 1)}{s^{n+1}}$
$\sin(at)$	$\frac{a}{s^2 + a^2}$
$\cos(at)$	$\frac{s}{s^2 + a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$



Find the Laplace transform of the following expression.

$$\mathcal{L}\{t^2 e^{-3t} \sin(5t)\} = ?$$

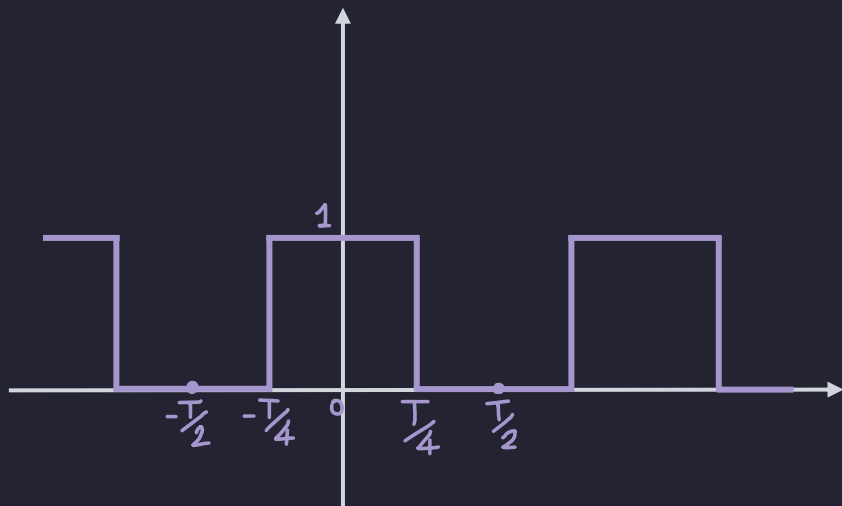


EXAMPLE 4,2 | Fourier Series

$$\tilde{f}(t) = \tilde{f}(t + T)$$

$$\sin\left(\frac{2n\pi}{T} \cdot t\right), \quad \cos\left(\frac{2n\pi}{T} \cdot t\right)$$

$$\frac{T}{n} = \text{دوره تناوب}, \quad \omega = \frac{2n\pi}{T}$$



$$\tilde{f}(t) = a_0 + \sum_{n=1}^{\infty} a_n \sin\left(\frac{2n\pi}{T} t\right) + b_n \cos\left(\frac{2n\pi}{T} t\right)$$

$$a_0 = \frac{1}{T} \int_0^T \tilde{f}(t) dt \quad (\text{مقدار متوسط})$$

$$a_n = \frac{2}{T} \int_0^T \tilde{f}(t) \sin(n\omega t) dt$$

$$b_n = \frac{2}{T} \int_0^T \tilde{f}(t) \cos(n\omega t) dt$$

$$\left\{ \begin{array}{l} a_0 = \frac{1}{2} \\ a_n = 0 \\ b_n = \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \\ \tilde{f}(t) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{2n\pi}{T} t\right) \end{array} \right.$$



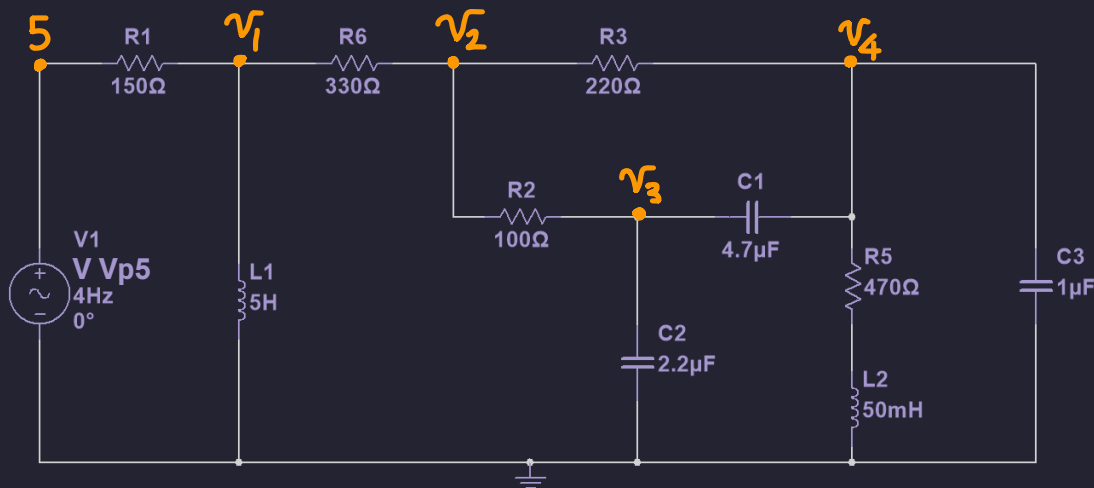
EXAMPLE 4,3 | Phasor Analysis and Complex Equation Systems

$$\frac{v_1 - 5}{150} + \frac{v_1}{Z_{L_1}} + \frac{v_1 - v_2}{330} = 0$$

$$\frac{v_2 - v_1}{330} + \frac{v_2 - v_3}{100} + \frac{v_2 - v_4}{220} = 0$$

$$\frac{v_3 - v_2}{100} + \frac{v_3 - v_4}{Z_{C_1}} + \frac{v_3}{Z_{C_2}} = 0$$

$$\frac{v_4 - v_2}{220} + \frac{v_4 - v_3}{Z_{C_1}} + \frac{v_4}{470 + Z_{L_2}} + \frac{v_4}{Z_{C_3}} = 0$$







The End!



THANKS!

Do you have any questions?



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